Gabriele Ciccolella



MSc Graduate in Intelligent Systems Engineering

AI & Machine Learning | Data Science | Quantum Computing & Cybersecurity

Rome, Italy | Date of birth: 13/06/2002
+39 328 026 6827 | gabriele.ciccolella02@gmail.com
Portfolio

PROFILE

MSc graduate in Intelligent Systems Engineering from Università Campus Bio-Medico di Roma. Focused on AI, Machine Learning and Data Science, with hands-on experience in deep learning, forecasting and predictive modelling. Complementary skills in AI agents, distributed systems, cybersecurity and quantum security.

EDUCATION

MSc in Intelligent Systems Engineering — **Rome, Italy**
Campus Bio-Medico University of Rome — 10/2024 - 10/2026
Final grade: 110/110 cum laude

BSc in Industrial Engineering — **Rome, Italy**
Campus Bio-Medico University of Rome — 09/2021 - 12/2024
Final grade: 104/110

TECHNICAL SKILLS

AI & Machine Learning: scikit-learn, LightGBM, Optuna, pandas and NumPy; supervised classification, feature engineering, hyperparameter optimisation, cross-validation and model evaluation.

Deep Learning & AI Agents: PyTorch, encoder-decoder LSTM, time-series forecasting; n8n, Ollama and Redis.

Cybersecurity: Ethical hacking and security assessment with Kali Linux, Wireshark, Nmap and Burp Suite; TCP/IP and ISO/OSI models; authentication, authorisation and hashing.

Quantum security: QKD/BB84, QBER estimation, finite-sample bounds, error reconciliation and Key Management System availability analysis.

Distributed Systems: Microservices, REST APIs, Docker, Docker Compose, Kubernetes, Prometheus and Grafana.

Robotics: Manipulator kinematics, geometric Jacobians, trajectory planning and simulation (academic training).

SELECTED PROJECTS

Quantum Key Distribution (QKD/BB84) — **MSc Thesis**
Developed an 8-microservice platform with Python, FastAPI and Docker Compose to simulate BB84, including error estimation, reconciliation and key verification.

Photovoltaic power forecasting — **Deep Learning**
Implemented a PyTorch encoder-decoder LSTM using 72 hours of history to forecast the next 24 hours with known future weather; feature engineering, early stopping and MAE/RMSE comparison against a baseline.

Machine Learning studies — **Academic Projects**
- Earthquake damage prediction with Random Forest/LightGBM, Optuna and nested cross-validation.
- Rehabilitation exercise recognition with 8 classifiers.
- KNN for oncology data.

AI Agent and Microservice Architecture — **Academic Project**
Built a conversational agent for requirements analysis with n8n, Ollama and Redis; integrated 6 Docker Compose services, metrics, dashboards and availability alerts.

Zanzo multiplayer card game | **Demo** — **Personal Project**
Python/JavaScript app for 2-8 players with PBKDF2 password hashing, token-based authentication, rate limiting, SQLite and WebSocket; automated tests and Cloudflare Workers deployment.

SKILLS & INTERESTS

Languages: Italian (native) | English (B2) | Spanish (basic)

IT Skills: Python, SQL, PyTorch, scikit-learn, pandas, NumPy, LightGBM and Optuna; FastAPI, Docker, Docker Compose, Git, pytest and GitHub Actions.

Office Skills: Microsoft Excel, Word and PowerPoint.

Interests: Chess (three-time university tournament finalist) | Competitive football (S.S. Lazio).

I hereby authorize the processing of my personal data for recruitment purposes in accordance with Regulation (EU) 2016/679 (GDPR) and applicable legislation.